Tensor computations

Border Comon's conjecture over the complex numbers

Difficulty: extreme **Importance:** interesting to the community

Statement

For every integer order $d \geq 3$, dimension $n \geq 2$, and symmetric tensor $T \in \mathrm{Sym}^d(\mathbb{C}^n)$, is

$$\underline{R}(T) = \underline{R}_{\mathrm{sym}}(T)?$$

Here $R(T)$ is the least number of summands in a decomposition $T = \sum_{j=1}^{r} v_{j,1} \otimes \cdots \otimes v_{j,d}$, with complex vectors. Symmetric rank restricts each summand to $c_j v_j^{\otimes d}$. The border rank $\underline{R}(T)$ is the least r for which a sequence of tensors of rank at most r converges to T in the entrywise Euclidean topology. Symmetric border rank instead uses symmetric tensors of symmetric rank at most r.

Relevance

This asks whether exploiting symmetry can increase the number of rank-one terms required for arbitrarily accurate approximation. It concerns the attainable accuracy of symmetric low-rank tensor models.

References

1. T. Mańdziuk and E. Ventura, *Symmetrization maps and minimal border rank Comon's conjecture*, arXiv:2411.05721v2, 2026. Full text, §1, opening discussion and Conjecture 1.1; the general conjecture precedes the minimal-border-rank specialization. Version record.
2. J. M. Landsberg, *Geometry and Complexity Theory*, Cambridge University Press, 2017. Author text, Conjecture 5.6.1.5, a three-factor formulation.

Status check — 2026-09-08

The September 3, 2026 revision of reference 1 describes the general conjecture as open and proves special minimal-border-rank cases. Searches *"border rank Comon" 2026 proof* and *"Comon" "border" "counterexample" "2025" OR "2026"* located no general resolution. The ordinary exact-rank Comon conjecture was disproved by Shitov; that counterexample does not settle this border-rank question. This is a bounded literature check, not a proof of absence of a solution.